

LDA Topic Modeling and Corpus Size

Topic modeling requires a large number of good-size documents for results reliability.

And **how large is large**? What would be considered the **least number of documents** (corpus size) for LDA/SLDA topic model?

Is a corpus of 200 documents large enough? Is 20 words a good-size document?

There are no magic numbers. The general rule of thumb is the more the better. Individual documents should also be of a decent size. Some researchers split documents into smaller units for analysis with interesting results.

Comparisons of documents in the hundreds and in the thousands yield significantly different results. Results based on thousands of documents are more reliable when evaluated by humans. In case of comparing different models (through topic coherence) it tells them apart to a good degree, whereas they tend to converge when the documents are in hundreds.

A good source of information for Gensim topic modeling is Micah Saxton's Capstone - Topic Modeling Best Practices.pdf (<https://msaxton.github.io/topic-model-best-practices/>).

Take-away lessons

- The **number of documents** is the most important variable; a few (e.g., tens of) documents won't work even if they are long; a few hundred documents should suffice but performance stabilizes after a large number of documents (1000 documents for 100 topics) (The figures quoted here are rules of thumb from one study, not thresholds the method imposes: a file is not a fixed quantity of text, and what counts is how much text there is and how it is divided.)
 - **Split large documents** into subdocuments if you have large but few documents
 - **Sample** a fraction of documents if you have too many documents
- **Document length** plays a useful role too; short documents (e.g., less than 10 words) won't work even if there are many of them; performance stabilizes after some large number of words per document (100 words for 100 topics)

How to tell whether YOUR topics are trustworthy

No document count can tell you this, and a corpus that clears every rule of thumb can still produce topics that do not hold. Two checks answer it directly, and both take minutes.

Run it twice. LDA starts from a random initialisation, so fit the model a second time and compare. If the same topics come back - the same words, grouped the same way - they are a property of your corpus. If they move about, they are a property of that particular run, and no

amount of interpretation will fix that. This is the single most useful check, and the one most often skipped.

Vary the number of topics. Ten is a default, not a finding. Fit for several values and see which topics survive: the ones that persist across settings are the solid ones. Topic coherence scores can guide the choice, but read the topics themselves - a coherent-looking score over word lists nobody can name is not a result.

If the topics are unstable, the usual remedy is more documents rather than more text: split long documents into chapters or sections so each becomes a document. This gives the model more separate contexts to compare, which is what per-document topic proportions are estimated from. The NLP Suite's File Splitter does it.

And treat topics as evidence to be read, not as output to be reported. A topic is a list of words that co-occur; what it MEANS is your reading of it, and that reading is part of the analysis rather than something the model supplies.